CPM: FLOAT OF AN ACTIVITY

Forty-eighth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



Float

- After computing earliest and latest times for all activities, the slack time for each activity has to be evaluated.
- Slack time is the length of time an activity can be delayed without delaying the entire project.
- **free float** or **free slack**.
- **Floats/slacks** are the slack times available within the allotted span of the non-critical activity. The two most common floats are the total float and the free float.

$$\text{Slack} = \text{LS} - \text{ES}$$

$$\text{or} \quad \text{Slack} = \text{LF} - \text{EF}$$

where

ES: Earliest start of an activity

LS: Latest start of an activity

EF: Earliest finish of an activity

LF: Latest finish of an activity

Slack Time and Identifying the Critical Path

- The activities with zero slack are called critical activities and are said to be on the critical path. The critical path is a continuous path through the project network that
  - Starts at the first activity in the project
  - Terminates at the last activity in the project
  - Includes only critical activities (i.e., activities with no slack time)

Critical Path Computation

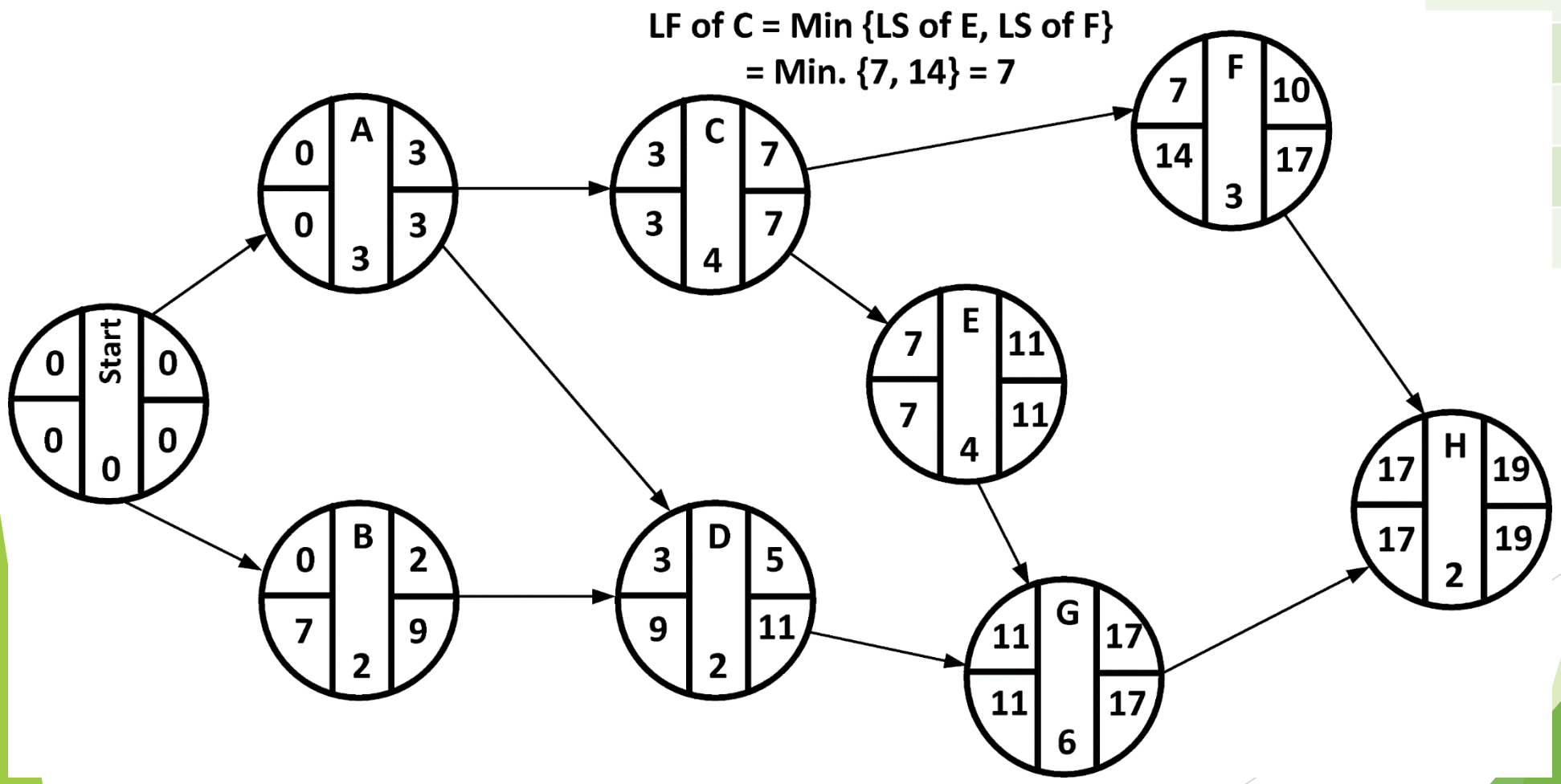
Computations are done for

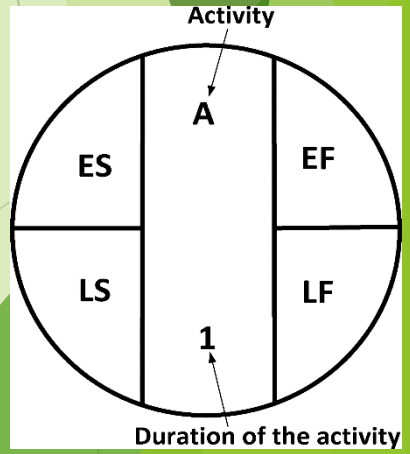
- Finding total duration needed to complete the project.
- Categorization of the activities as critical and non-critical.

- An activity is said to be critical, when there is no “leeway” in determining its start and finish time.
- A non-critical activity shows some scheduling slack.

- ***Example:*** Calculate the slack time for the activities as per the information provided.

Activities	Predecessor (s)	Duration (weeks)
A	--	3
B	--	2
C	A	4
D	A, B	2
E	C	4
F	C	3
G	D, E	6
H	F, G	2





Activity	Activity Time	Earliest start (ES)	Earliest finish (EF)	Latest start (LS)	Latest finish (LF)	Slack (LS–ES) or (LF–EF)	On Critical Path
A	3	0	3	0	3	0	Yes
B	2	0	2	7	9	7	No
C	4	3	7	3	7	0	Yes
D	2	3	5	9	11	6	No
E	4	7	11	7	11	0	Yes
F	3	7	10	14	17	7	No
G	6	11	17	11	17	0	Yes
H	2	17	19	17	19	0	Yes

The ES, EF, LS, LF and slack time of all activities are shown in Table.

Activity B has slack time of 7 week. Its LS is 7 and ES is 0; its LF is 9 and EF is 2.

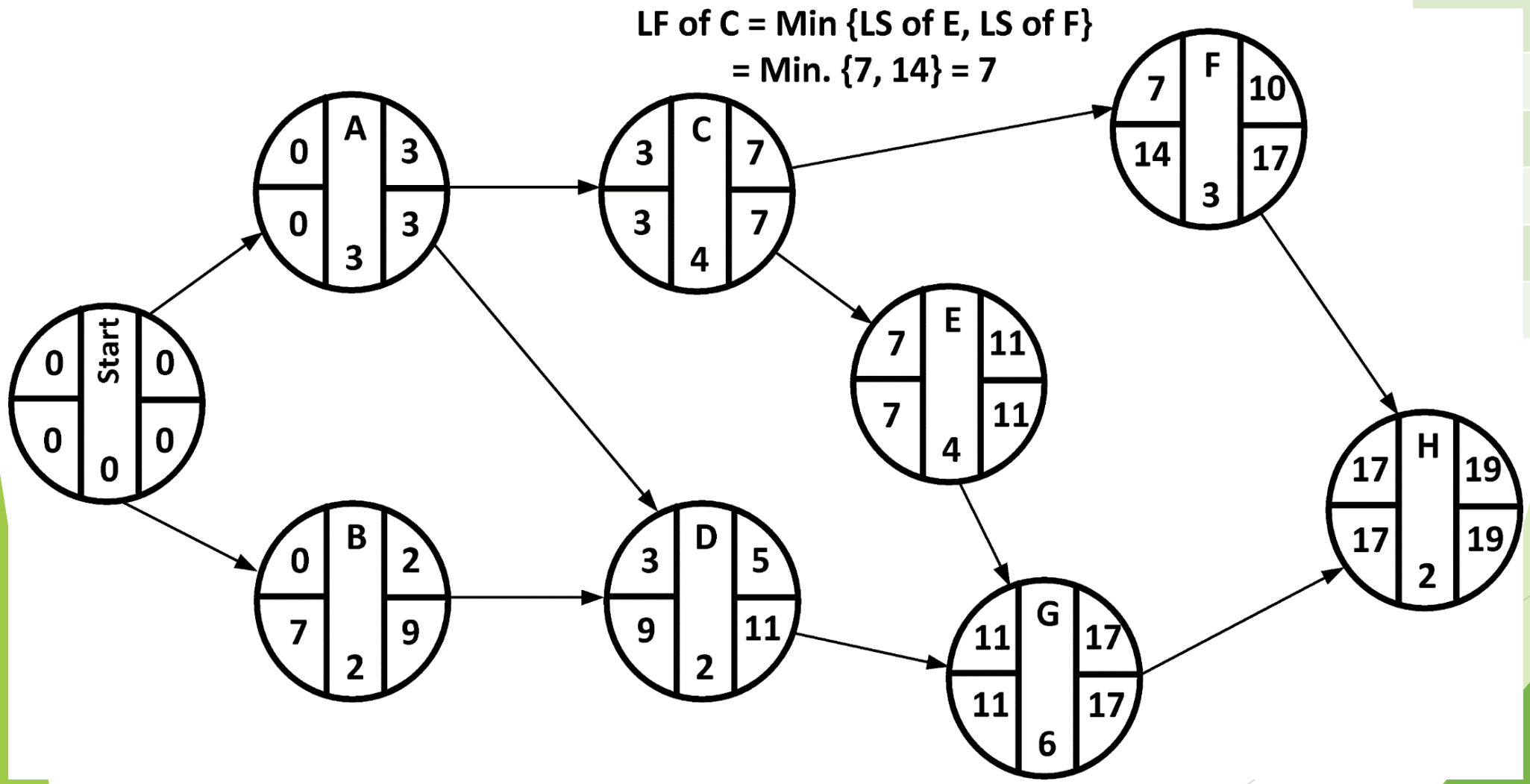
It implies that activity B can be delayed up to 7 weeks while the project be still completed in 19 weeks.

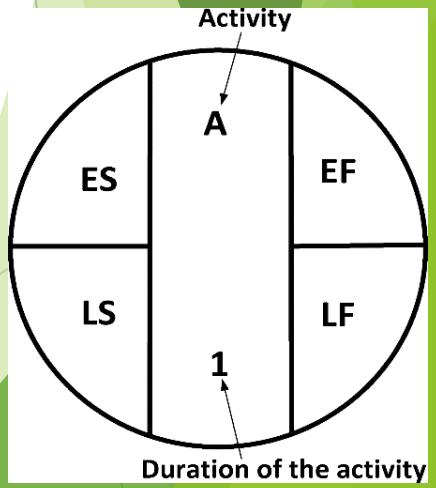
Activity	Activity Time	Earliest start (ES)	Earliest finish (EF)	Latest start (LS)	Latest finish (LF)	Slack (LS–ES) or (LF–EF)	On Critical Path
A	3	0	3	0	3	0	Yes
B	2	0	2	7	9	7	No
C	4	3	7	3	7	0	Yes
D	2	3	5	9	11	6	No
E	4	7	11	7	11	0	Yes
F	3	7	10	14	17	7	No
G	6	11	17	11	17	0	Yes
H	2	17	19	17	19	0	Yes

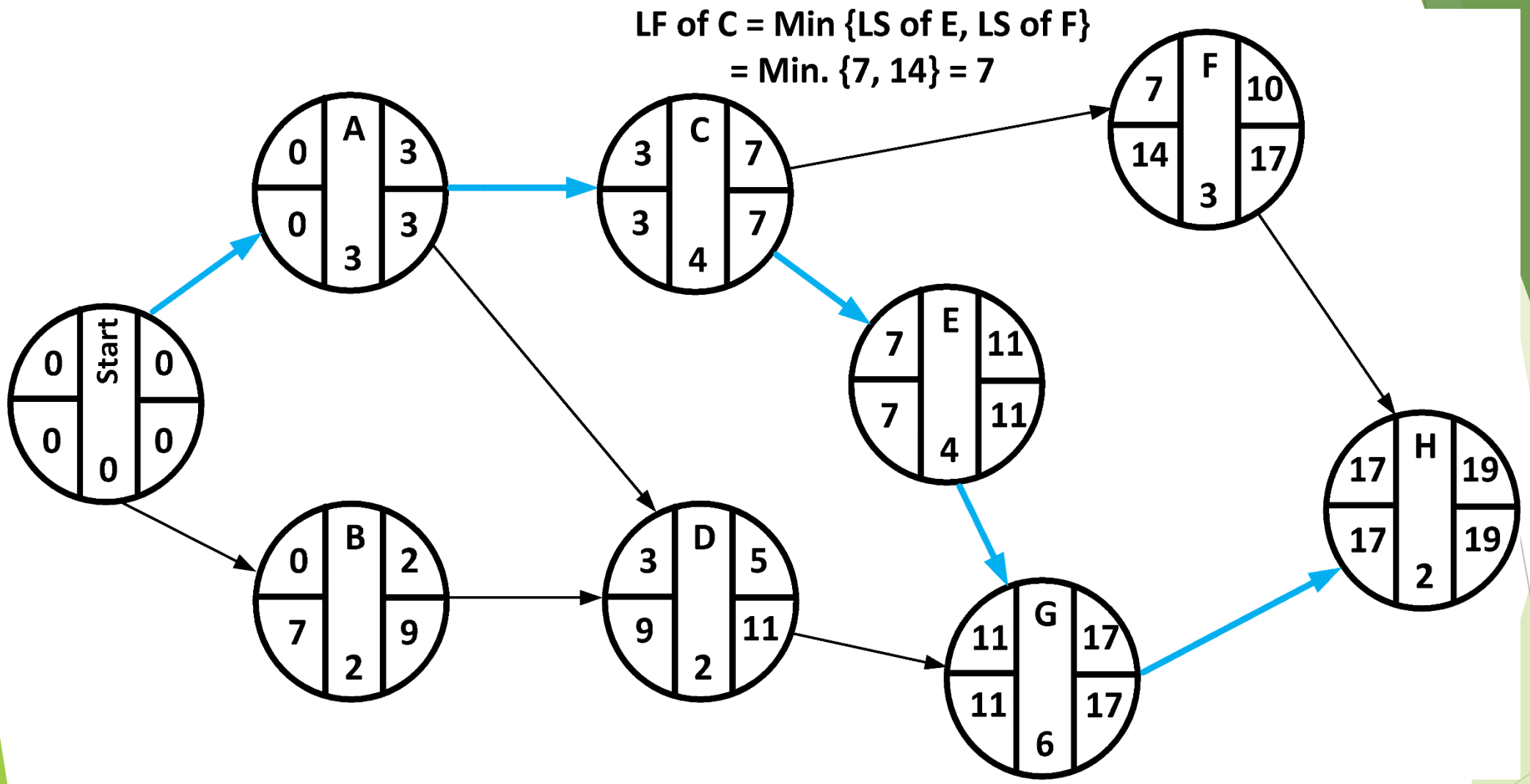
Other activities (A, C, E, G, H) have no slack time. In such cases, delay of any activity will delay the entire project.

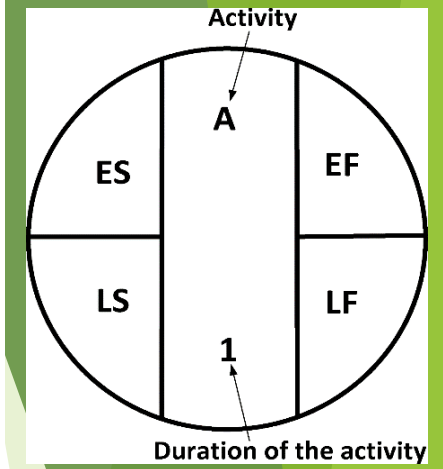
- ***Example:*** Show the critical path of the project. Find the project completion time.

Activities	Predecessor (s)	Duration (weeks)
A	--	3
B	--	2
C	A	4
D	A, B	2
E	C	4
F	C	3
G	D, E	6
H	F, G	2

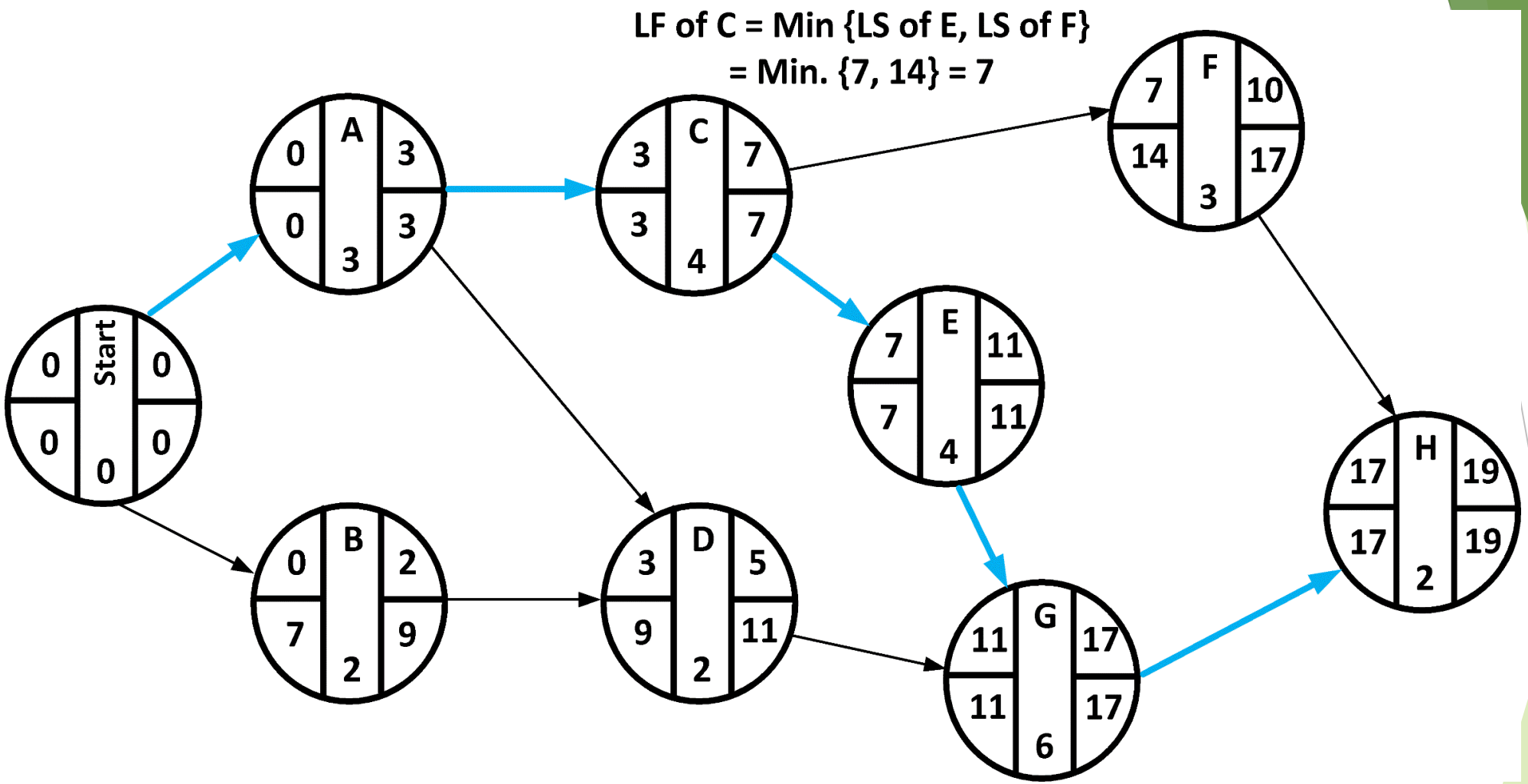


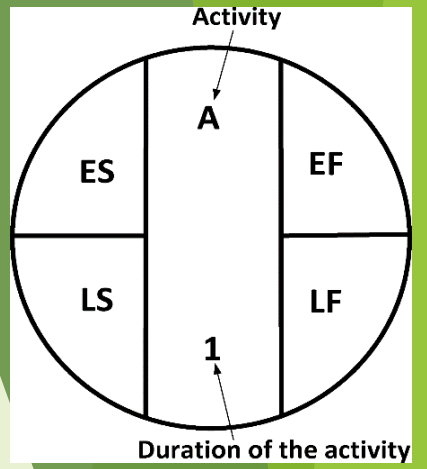






- The project completion time is 19 weeks.
- The critical path shown by **blue line** is Start–A–C–E–G–H.





- The critical path follows the activities with zero slack. This path (Start–A–C–E–G–H) is the longest in the network.
- Activities B, D and F are not a part of critical path as they have slack of 7, 6 and 7 weeks, respectively.

Table shows the calculation of slack for each activity

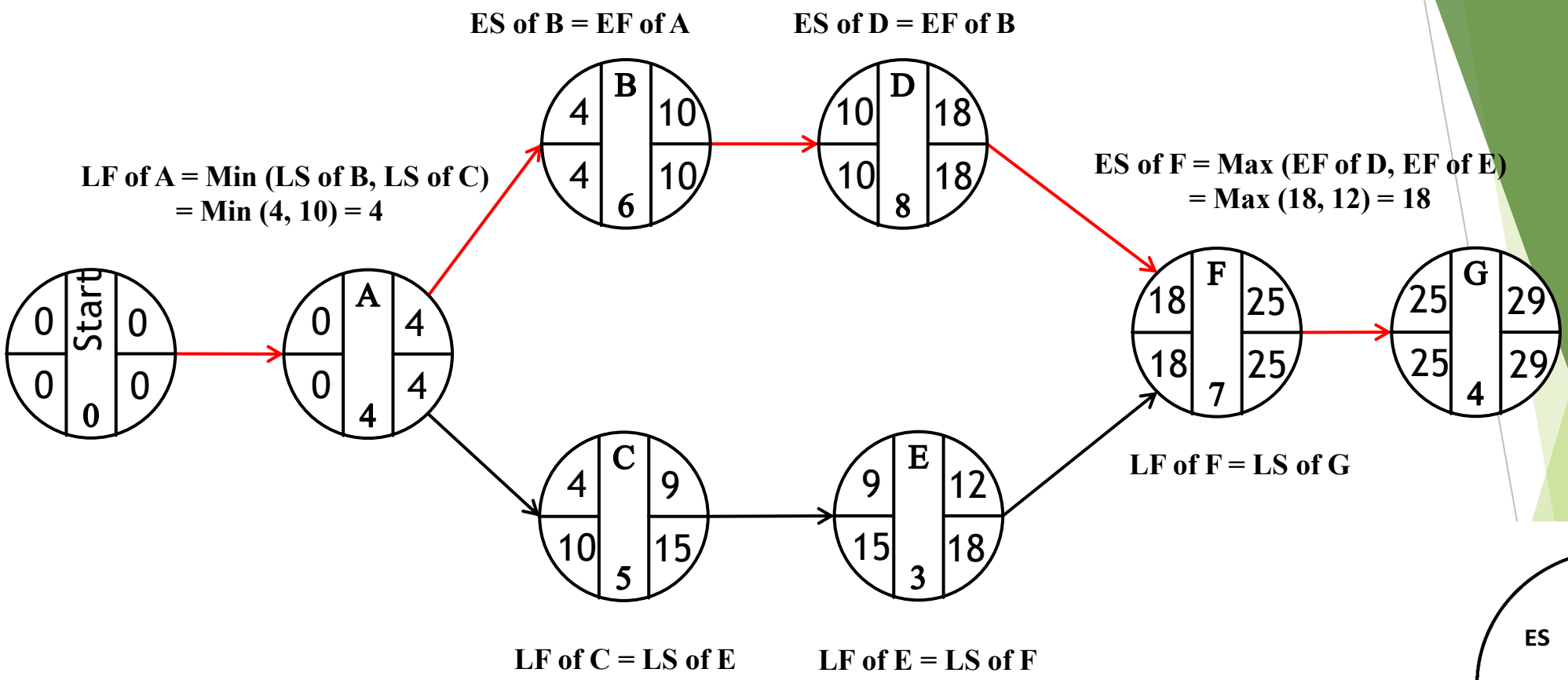
Activity	Duration	ES	EF	LS	LF	Slack = LF – EF = LS – ES
A	3	0	3	0	3	0
B	2	0	2	7	9	7
C	4	3	7	3	7	0
D	2	3	5	9	11	6
E	4	7	11	7	11	0
F	3	7	10	14	17	7
G	6	11	17	11	17	0
H	2	17	19	17	19	0

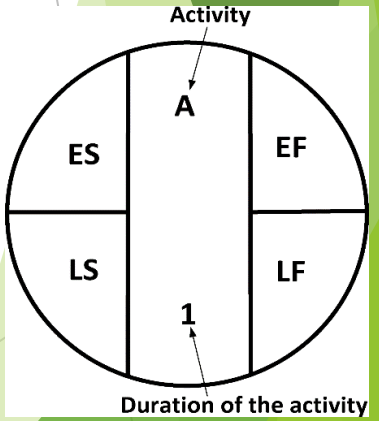
Total Slack time

- Delaying activity B causes both activities B and D to lose their slack. This type of slack is referred to as **total slack**.
- When two or more non-critical activities successively appear in a path, they share the total slack.

Example: It is planned to launch an electric car in the market. Total 7 activities are needed as per the information provided. Find out the total and free floats of the activities.

Activity	Description	Predecessor
A	Market survey	----------
B	Design	A
C	Market strategy	A
D	Design of production facility	B
E	Selection of advertising method	C
F	Prototype testing	D, E
G	Launching	F





Critical path is shown in red colour.

The critical path of the project is: **A–B–D–F–G** and critical activities are **A**, **B**, **D**, **F** and **G**.

Activity i	Duration	ES_i	ES_j	EF_i	LS_i	LF_i	Free float $= ES_j - EF_i$	Total float $= LF_i - ES_i$	Critical activity?
A	4	0	4	4	0	4	0	0	Yes
B	6	4	10	10	4	10	0	0	Yes
C	5	4	9	9	10	15	0	6	No
D	8	10	18	18	10	18	0	0	Yes
E	3	9	18	12	15	18	6	6	No
F	7	18	25	25	18	25	0	0	Yes
G	4	25	29	29	25	29	0	0	Yes

i : current activity

j : succeeding activity

ES: Earliest start of an activity

LS: Latest start of an activity

EF: Earliest finish of an activity

LF: Latest finish of an activity

Important

- Total float of activity (I, J)=Latest occurrence of event J– Earliest occurrence of event I –Activity duration
- Free float of activity (I, J)=Earliest occurrence of event J– Earliest occurrence of event I –Activity duration
- Free float is always less than or equal to total float

Red-Flagging rule

- If free float is equal to total float, then activity can be scheduled anywhere within the span (earliest occurrence of event I, latest occurrence of event j).
- If free float is less than total float, then the start of the activity can be delayed by no more than the free float relative to its earliest start time without causing schedule conflict. Any delay larger than free float must be accompanied by the corresponding delay of all the activities immediately succeeding it.